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Appearance and ionization energies of C_{60-2m}^{z+} and C_{70-2m}^{z+} ions (with z and m up to 4) produced by electron impact ionization of C_{60} and C_{70} , respectively

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Electron impact ionization of C_{60} and C_{70} has been studied as a function of electron energy in the threshold region using a crossed molecular beam/electron beam double focusing sector field mass spectrometer system. Appearance energies have been determined (using a special threshold extrapolating procedure) for C_{60-2m}^{z+} and C_{70-2m}^{z+} ions with z ranging from 1 to 4 and m from 0 to 4, respectively. Whereas the different charge states of the parent ions of C_{60} and C_{70} have the same appearance energies within the experimental error bars, the fragment ions of C_{70} show slightly higher values than their C_{60} counterparts due to an increased kinetic shift. The presently deduced ionization energies for the C_{60} parent ions are in excellent agreement with previously reported reliable data for C_{60}^{z+} ions (previous data are available only up to z is 3). In contrast, these experimental values disagree strongly with a theoretical calculation by Yannouleas and Landman predicting an approximately linear dependence of the ionization energy on the charge state z , i.e., the experimental values deviate strongly from linearity with increasing charge.

I. INTRODUCTION

The number of scientists working with carbon fullerenes increased dramatically since the discovery of the production of C_{60} in macroscopic amounts in 1990.¹ Many exciting properties of this novel class of molecules/clusters were discovered and soon also the existence of multiply charged C_{60}^{z+} ions (recently up to $z=7$) was reported in mass spectra and their production was investigated.²⁻²⁰ For singly and doubly charged C_{60} ions ionization energies were measured with different methods and the values of 7.64 for C_{60} (Ref. 8) and of 11.43 for C_{60}^+ (Ref. 11) are well established today. For C_{60}^{2+} the reliable (see discussion in Ref. 17) ionization energies determined so far include the charge stripping result of 17.0 eV by Lifshitz *et al.*,¹² the corrected charge transfer bracketing results of 15.7 and 15.6 eV by McElvany *et al.*³ and Javahery *et al.*,⁵ respectively, and our own electron impact ionization result¹⁷ of 16.6 eV. Moreover, Scheier *et al.*¹⁷ have recently measured the appearance energies for singly-, doubly-, and triply-charged C_{60} fragment ions and compared these data with results for singly-charged fragment ions reported by Anderson and co-workers²¹ using a charge and energy transfer fragmentation technique.

In extending these previous studies we report here (i) a complete set of data for singly-, doubly-, and triply-charged parent and fragment ions for C_{70} and (ii) for the first time appearance energies for quadruply charged fullerene ions including the parent ions C_{60}^{4+} , C_{70}^{4+} and the fragment ions C_{58}^{4+} , C_{56}^{4+} from C_{60} and C_{68}^{4+} , C_{66}^{4+} , C_{64}^{4+} and C_{62}^{4+} from C_{70} , respectively. As (i) all the quadruply-charged fullerene ions may overlap with doubly-charged carbon cluster ions and as (ii) the even numbered quadruply-charged ions may in addition coincide with singly-charged carbon cluster ions, some of the present appearance energy measurements had to be made using the respective isotopomer ion containing one ^{13}C atom in order to exclude possible coincidences with contaminating lower-charged ions. Whereas the different charge states of

the parent ions of C_{60} and C_{70} have the same appearance energies within the experimental error bars, the fragment ions of C_{70} show slightly higher values than their C_{60} counterparts due to an increased kinetic shift (i.e., see Refs. 22 and 23) in the case of the larger fullerene parent ion.

The presently deduced ionization energies for the C_{60} ions are in excellent agreement with previously reported reliable data for C_{60}^{z+} ions (previous data are available only up to z is 3). In contrast, these experimental values disagree strongly with a theoretical calculation by Yannouleas and Landman²⁴ predicting an approximately linear dependence of the ionization energy on the charge state z (see also an earlier similar prediction for aromatic hydrocarbons by Smith²⁵), i.e., the experimental values deviate strongly from linearity with increasing charge.

II. EXPERIMENT

The mass spectrometer system used in this experiment consists of a double focusing sector field mass spectrometer (reversed geometry) with a modified open electrode²⁶ Nier-type electron impact ion source. The mass resolution for the present measurements was set to approximately 1000 in order to clearly distinguish isotopic peaks containing one ^{13}C atom from peaks containing only ^{12}C atoms (see Fig. 1). The electron current for the ionization cross section measurements was set at a maximum of 50 μA in order to prevent space charge effects in the ion source and extraction region and multiple electron collisions during the ionization process (see the discussion in Ref. 20). The electron energy scale was calibrated with help of measured ionization efficiency curves of singly and doubly charged Ar or Ne ions using z th root extrapolation to the respective appearance energies.²⁷ Moreover, the potentials of the lenses in the ion source and ion extraction region were tuned in such a way as to reproduce the known ionization efficiency curves of Ar or Ne ions.²⁶ This procedure helps according to Ref. 26 to measure the

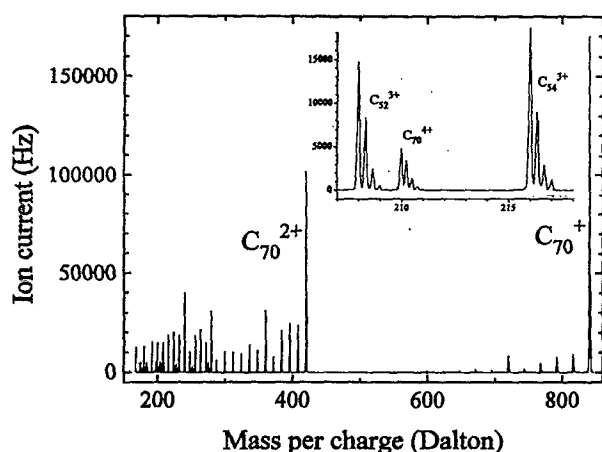


FIG. 1. Mass spectrum (semilog scale) of C₇₀ using 200 eV electrons. Insert: Isotope-resolved C₇₀⁴⁺ peak structure exhibiting the first four isotopomers of this ion, i.e. ¹²C₇₀⁴⁺, ¹³C¹²C₆₉⁴⁺, ¹³C¹²C₆₈⁴⁺, and ¹³C¹²C₆₇⁴⁺, respectively.

accurate shape of partial ionization cross section curves from which meaningful ionization energies can be deduced. The C₆₀ and C₇₀ samples, respectively, were evaporated at a temperature of 890 K in a temperature-controlled oven close to the ion source thereby forming an effusive beam entering the ionization region through a large orifice in the ion source housing. The purity of the C₆₀ powder used was 99.99% and that of C₇₀ 98%, respectively.

III. RESULTS

For C₇₀ a whole series of ionization efficiency curves were measured for all singly- and multiply-charged (up to quadruply-charged) parent and fragment ions down to C₆₂^{z+}. It was not possible to measure C₆₀ as a fragment ion from C₇₀, because (i) the main impurity in the C₇₀ sample is C₆₀ and (ii) in addition small traces of C₆₀ came from the vessel walls as a residual background and memory effect from earlier measurements. Table I gives all appearance energies determined from the measured ionization curves of these ions produced by electron impact of the neutral C₇₀ vapor. Also included are the results of our earlier measurements on singly-, doubly-, and triply-charged parent and fragment ions of C₆₀ (Ref. 17) and the present results on quadruply-charged parent and fragment ions of C₆₀.

As the ionization efficiency curves for fullerene ions—even for the singly charged parent ions (Fig. 2 gives the respective data for C₆₀ and C₇₀ parent ions)—are strongly curved close to the onset, it is rather difficult to determine the appearance energy in the usual way by extrapolating the data points back to the *x*-axis [see, for example, the ionization curves shown on the left-hand side of Figs. 2(a) and 2(b), respectively]. Following a recently proposed extrapolation procedure¹⁷ we raised the original data points to a certain power (1/*p*) in order to linearize the lower part of the ionization efficiency curve over at least 10 eV [see the linearized ionization curves shown on the right-hand side of Figs. 2(a) and 2(b), respectively]. For these fullerenes ions the power (*p*) had to be chosen considerably larger than in the case of atomic ions which follow a *z*th power law with the charge state *z*.²⁷ The reason for this is the large number of electronic and vibrational states of the fullerene ions which can be populated by electron impact ionization. Each state itself should obey the power law of the charge but the sum over all of these states gives an additional curvature. This strongly curved onset of the ionization efficiency curve not only exists in the case of electron impact ionization but also limits the accuracy in the determination of appearance energies via photon-ionization of C₆₀. For more information about the present evaluation method see Ref. 17.

Figure 3 shows the appearance energies of the parent and fragment ions of C₇₀ (open symbols) compared with those of C₆₀ (filled symbols). The appearance energies of the fragment ions of both fullerenes exhibit a remarkably linear relationship with the number of lost C₂ units. A similar result (with the exception of C₅₂⁺) has been reported by Anderson and co-workers²¹ for appearance energies for singly-charged C₆₀ fragment ions obtained by studying collisions between Ne⁺ and neutral C₆₀ as a function of ion energy. These results indicate that the energy necessary to remove an additional C₂ unit from a fragment ion is approximately constant. As this energy is also independent from the charge state (all curves have the same slope) we conclude that there is a common underlying statistical process for the evaporation of a C₂ unit independent of the charge state or precursor size of the dissociating ion once the first fragment ion has been produced. Moreover, it can be also stated that the formation of these fragment ions of C₆₀ and C₇₀ is (at least in the onset region²⁸) caused by sequential evaporation of single C₂ units rather than by a single fissioning process where the differing binding energies of larger carbon fragments should be visible

TABLE I. Appearance energies (in eV) for singly-, doubly-, triply-, and quadruply-charged parent and fragment ions of C₆₀ and C₇₀ ions, respectively. The values for the singly-, doubly-, and triply-charged parent and fragment ions of C₆₀ are from Ref. 17.

<i>m</i>	C _{60-2m} ^{z+}				C _{70-2m} ^{z+}			
	<i>z</i> =1	<i>z</i> =2	<i>z</i> =3	<i>z</i> =4	<i>z</i> =1	<i>z</i> =2	<i>z</i> =3	<i>z</i> =4
0	7.6±0.5	19 ±0.5	35.6±1	63±2	8 ±0.5	19.4±0.7	34.9±1.5	62.5±2
1	43.7±1.5	54 ±2	70.3±2.2	92±3.5	49.4±1.4	61.5±1.6	73.9±2.3	94 ±3.8
2	48.8±1.6	59.8±1.9	76 ±2	95±4.4	54 ±1.7	66.1±2	79.9±2.6	100 ±4.1
3	54.3±2	65.1±2.4	82.2±3.1	108±5	59.5±1.9	71.8±2.8	85.2±2.9	105 ±5.2
4	59.9±2.1	71.8±3.3	90.6±4.1		64.6±2.2	76.2±3.5	90.5±3.9	112 ±4.8

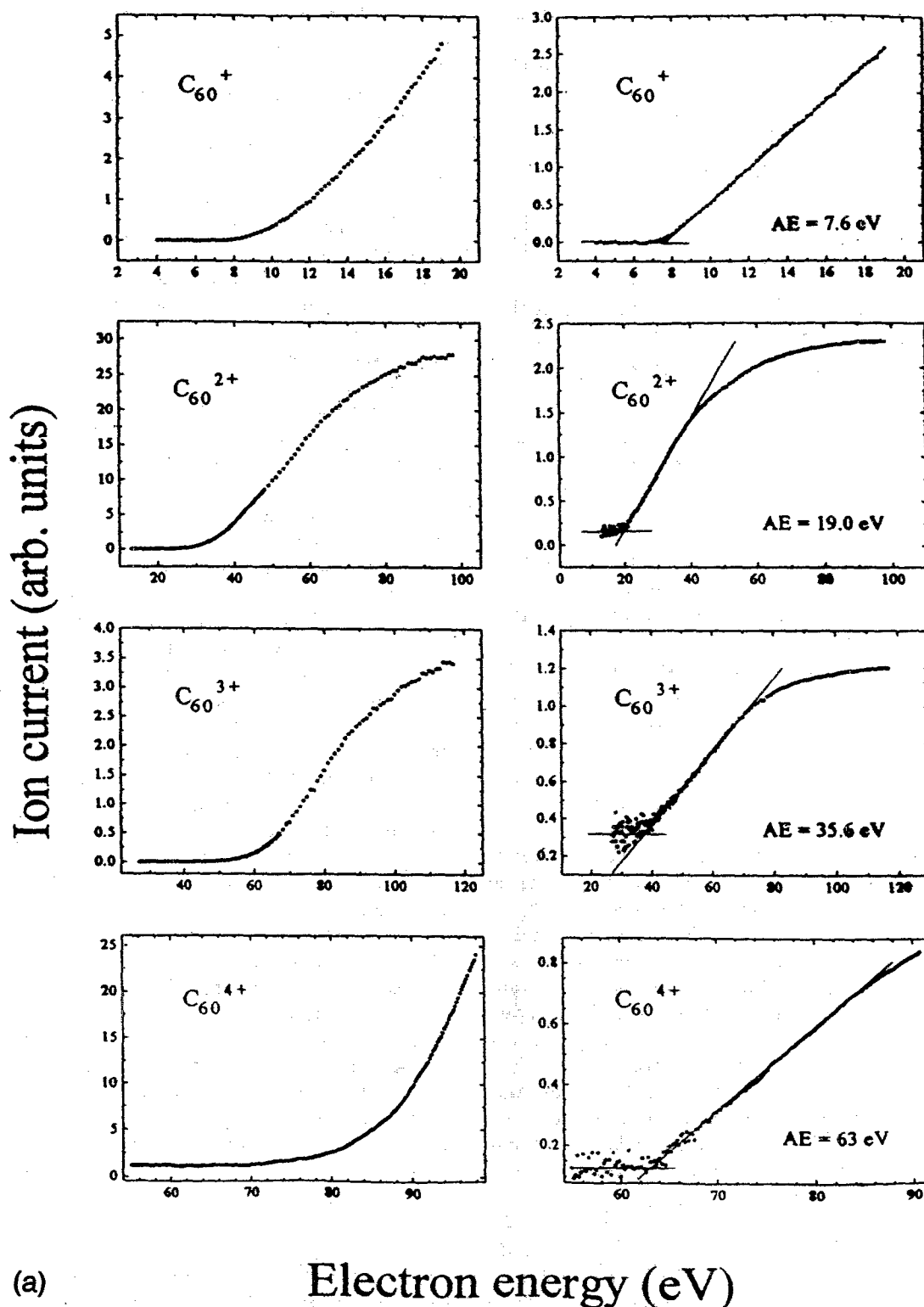


FIG. 2. (a) Ionization efficiency curves for the electron impact induced production of C_{60}^{+} , C_{60}^{2+} , C_{60}^{3+} , and C_{60}^{4+} (left-hand side) and the same data sets modified by a power law (with an exponent p of 1.67, 4, 6.7, and 10, respectively) in order to linearize the threshold region (right-hand side). The lines shown represent the fitted linear regression allowing an easy determination (see text) of the appearance energy AE. (b) Ionization efficiency curves for the electron impact induced production of C_{70}^{+} , C_{70}^{2+} , C_{70}^{3+} , and C_{70}^{4+} (left-hand side) and the same data sets modified by a power law (with an exponent p of 1.7, 4, 7, and 10, respectively) in order to linearize the threshold (right-hand side). The lines shown represent the fitted linear regression allowing an easy determination (see text) of the appearance energy AE.

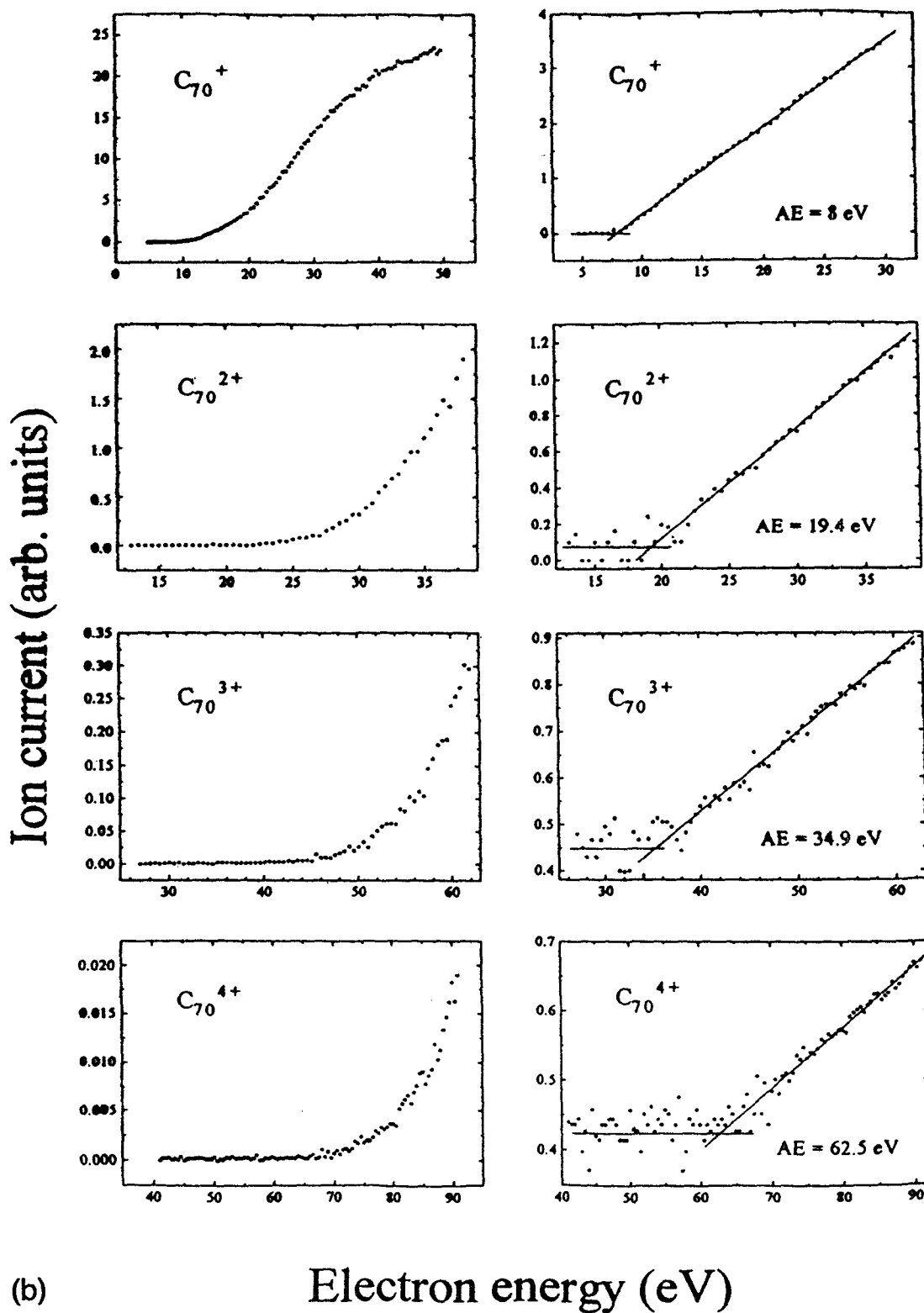


FIG. 2. (Continued.)

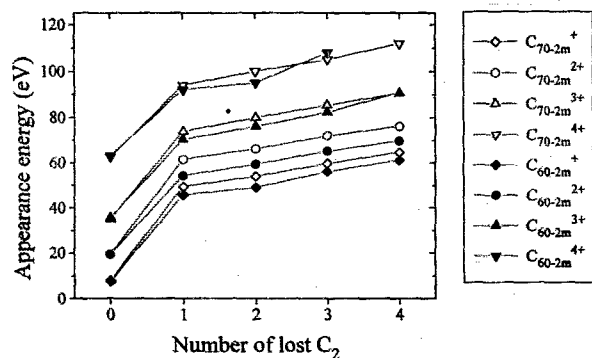


FIG. 3. Appearance energies of singly-, doubly-, triply-, and quadruply-charged ions produced by electron impact ionization of C_{70} (open symbols) and of C_{60} (filled symbols) as a function of the number of C_2 units lost in the ionization process.

as irregularities in the curves shown in Fig. 3.

Moreover, whereas the additional energy to remove a further C_2 unit from a fragment ion is of the order of 5.7 ± 1.3 eV (and may be related to the relative binding energies via statistical models, see Ref. 23), the energy to remove such a C_2 unit from the parent ion is in the case of C_{60} approximately 36 eV. This huge kinetic shift is caused by the many degrees of freedom and the rather large binding energy $BE(C_{58}^+-C_2)$ of $7.1(\pm 0.4)$ eV determined recently by Foltin *et al.*²² from the electron impact cross section threshold behavior of the first fragment ions using RRKM and EEM theory. In this context it is interesting to note that the appearance energy of all the fragment ions of C_{70} lies consistently above the appearance energy of the corresponding fragment ions of C_{60} losing the same number of C_2 constituents (see Fig. 3). In contrast the appearance energies for the mother ions C_{60}^{z+} and C_{70}^{z+} are more or less the same within the error limits. Although the difference between the appearance energies of the corresponding fragment ions of C_{60} and C_{70} in most of the data points is smaller than the uncertainty of the values, the general trend that the appearance energy points of the C_{70} fragment ions lie above those of the C_{60} fragment ions is statistically significant, because it occurs for all charge states and almost all fragment ion sizes. The reason for this difference can be traced to the much larger number of degrees of freedom of C_{70} , thereby causing a larger kinetic shift in the appearance energy of the fragment ions. Therefore, assuming similar binding energies in the same time regime and with the same amount of internal energy, C_{70} should be more stable than C_{60} against neutral C_2 evaporation.

The overriding result and conclusion in the present study concerns, however, the behavior of the ionization energy of the parent fullerene ions vs final charge state. In this paper the appearance energies of C_{60}^{4+} and C_{70}^{4+} and thereby the ionization energies of C_{60}^{3+} and C_{70}^{3+} were measured for the first time, thus allowing a much better appraisal of this relationship (i.e., over a wider z range) than in the past having available experimental data only up to triply-charged C_{60} . Recent theoretical investigations by Yannouleas and

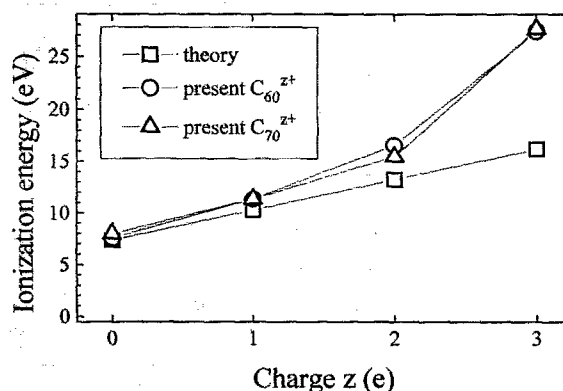


FIG. 4. Ionization energy vs initial charge state z for C_{60} (open circles) and C_{70} (open triangles) parent ions derived from the respective appearance energies given in Table I. Also shown for comparison theoretical predictions (open squares) by Yannouleas and Landman (Ref. 24) using a new LDA method.

Landman²⁴ based on a new local density functional method including (i) a stabilized jellium approximation; (ii) a shell correction method; and (iii) the effect of icosahedral symmetry, predict that the ionization energy for C_{60} should exhibit a linear relationship with the charge state z (similar predictions have been made for large aromatic hydrocarbons, see Smith²⁵). Figure 4 shows the ionization energies for C_{60}^{z+} and C_{70}^{z+} (with z from 1 to 4) vs z . The open squares are the values predicted by Landman and co-workers for C_{60}^{z+} and the open circles and triangles are the experimental determined values of the present work for C_{60}^{z+} and C_{70}^{z+} , respectively. From this comparison it clearly follows that the present experimental data exhibit a nonlinear relationship with z in contrast to the theoretical prediction. In order to strengthen this finding it is important to note that the present data for C_{60} are in very good agreement with previous reliable determinations (using photoionization,⁷⁻¹¹ charge stripping,^{12,13} and corrected charge transfer⁵ techniques) in the available range up z is 3.

In the previous discussion on this matter the theoretical findings could only be compared to data including the first three ionization energies. Based on this comparison Landman and co-workers came to the conclusion that their theoretical model agrees quite well with the experimental data. This was, however, mainly caused by the fact that the deviation between the experimental and theoretical data begins to show up only very slightly at the second and third ionization energy and that Landman and co-workers compared their second and third ionization energy with early charge transfer values which were not yet corrected for Coulomb repulsion (see discussion in Refs. 5,6) and with appearance energies for C_{60}^{2+} and C_{60}^{3+} obtained by Baba *et al.*¹⁶ with a method,³⁰ which gives due to its experimental technique and its data analysis used, results at variance with the other reliable methods (see discussion in Ref. 17). In contrast, the present method has been carefully calibrated with other systems and against other independent methods. In addition, for the triply-charged fullerenes, the difference between the predicted theoretical value and the experimental determined ion-

ization energy is more than 11.5 eV, which is far beyond any conceivable experimental error. Nevertheless, further experimental and theoretical work with upgraded methods is certainly recommended in order to settle this issue of the true behavior of this very interesting and fundamental relationship between energetics and charge state of successively charged molecular compounds.

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